

DEVICE RISK ANALYSIS: FINAL PROJECT REPORT

Machine Learning Case Study No. 86 | Department of CSE

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Executive Abstract

This project report presents an end-to-end Machine Learning classification system designed to analyze multi-dimensional device telemetry and predict endpoint security risk tiers: Low Risk, Medium Risk, and High Risk. The system evaluates 3,012 device instances across 17 security attributes. Following structured data quality audits, deduplication, and median/mode imputation, six classification algorithms were trained and benchmarked. Tuned Logistic Regression achieved champion performance with 89.00% test accuracy, an F1-Score of 0.8898, and an exceptional multi-class ROC-AUC of 0.9756. A production-ready Streamlit web application provides real-time on-demand inference.

1. Algorithm Benchmark & Comparative Performance

Algorithm	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression (Tuned)	0.8900	0.8900	0.8900	0.8898	0.9756
Support Vector Machine	0.8800	0.8802	0.8800	0.8801	0.9710
Gradient Boosting (Tuned)	0.8500	0.8521	0.8500	0.8502	0.9580
Random Forest Classifier	0.8017	0.8101	0.8017	0.8021	0.9265
K-Nearest Neighbors	0.7883	0.7908	0.7883	0.7868	0.9151
Decision Tree Classifier	0.7167	0.7327	0.7167	0.7178	0.8466

2. Top 5 Security Risk Drivers

Rank	Factor	Weight	Operational Impact
1	Operating System	17.0%	Kernel architecture & update channel compliance
2	Patch Status	11.4%	Latency in applying vendor security hotfixes
3	Vulnerability Count	9.6%	Active unmitigated software CVEs detected
4	Suspicious Activity	7.6%	Network IDS behavioral anomaly events flagged
5	Compliance Score	6.9%	Engineered composite score of defense settings

3. Academic Conclusion

The project demonstrates how supervised machine learning transforms endpoint security operations by providing objective, repeatable, and fast risk classification.